A 21-COLORING OF THE PLANE WITHOUT MONOCHROMATIC UNIT-AREA RECTANGLES

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ABSTRACT. Erdős and Graham asked whether every finite coloring of the plane must contain a monochromatic rectangle of any prescribed area. Kovač gave a negative answer by constructing a 25-coloring with no monochromatic rectangle of area 1. We reduce the number of colors to 21 by replacing the square cells in his construction with regular hexagons.

1. INTRODUCTION

Euclidean Ramsey theory asks which finite point configurations must occur monochromatically in every finite coloring of Euclidean space. In 1980, Graham proved that every finite coloring of $\mathbb{R}^n$ contains a monochromatic n-simplex with any prescribed positive volume [2]. Erdős and Graham asked whether an analogous statement holds for rectangles and parallelograms [1]. The rectangle problem was still open in the 2015 second edition of Graham and Butler's *Rudiments of Ramsey Theory* [3].

In 2026, Kovač answered the rectangle question by constructing a 25-coloring of $\mathbb{R}^2$ with no monochromatic rectangle of area 1 [4]. For rectangles, scaling shows that it suffices to consider only rectangles of area 1, so this answers the question negatively. His coloring has a stronger property: it contains no monochromatic parallelogram whose two adjacent side lengths have product 1. We improve the number of colors from 25 to 21.

Theorem 1. *There exists a* 21*-coloring of* $\mathbb{R}^2$ *with no monochromatic parallelogram whose two adjacent side lengths have product* 1*.*

For a rectangle, the product of two adjacent side lengths is its area, so the theorem gives a 21-coloring with no monochromatic rectangle of area 1. The analogous question for parallelograms of prescribed area is more tricky and remains open [4]. If u and v are adjacent side vectors and θ is the angle between them, then the area is $|u||v||\sin\theta|$. The theorem rules out only the case $|u||v| = 1$, and a parallelogram can have area 1 with $|u||v| > 1$; for instance, take $\sin\theta = 1/(|u||v|)$.

Our construction follows Kovač's approach. We identify the plane with $\mathbb{C}$, assign colors according to the location of z^2, and replace his periodic square tiling with the Voronoi tiling of the triangular lattice. The relevant sublattice has index 21. Section 2 gives the coloring and proves Theorem 1; Section 3 shows that no smaller index works in this construction.

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2. Proof of Theorem 1

Identify $\mathbb{R}^2$ with $\mathbb{C}$. Let $P = ABCD$ be a possibly degenerate parallelogram. We write its vertices as $z_A = z$, $z_B = z + u$, $z_C = z + u + v$, and $z_D = z + v$.

Consider the alternating square sum

$$\mathscr{I}(P) = z_A^2 - z_B^2 + z_C^2 - z_D^2. \tag{2.1}$$

Substituting the vertex coordinates gives $\mathscr{I}(P) = z^2 - (z+u)^2 + (z+u+v)^2 - (z+v)^2 = 2uv$. Thus $|\mathscr{I}(P)| = 2$ whenever the product of the two adjacent side lengths is 1. We will therefore construct a coloring for which $\mathscr{I}(P)$ avoids the circle $|w| = 2$ whenever P is monochromatic.

Let $\omega = e^{\pi i/3}$ and $L = \mathbb{Z} + \mathbb{Z}\omega$. The Voronoi cell H_0 of 0 in L is a regular hexagon with inradius $1/2$ and circumradius $1/\sqrt{3}$. Choose a half-open version V of H_0 by assigning each boundary point of the Voronoi tiling to one adjacent cell. Then the translates $\lambda + V$, with $\lambda \in L$, partition $\mathbb{C}$. Set $\alpha = 4 + \omega$.

For $j \in \{0, 1, \dots, 20\}$, define

$$\mathscr{C}_j = \Big\{ z \in \mathbb{C} : z^2 \in \frac{5}{6}(p + q\omega + V) \text{ for some } p, q \in \mathbb{Z} \text{ with } p - 4q \equiv j \pmod{21} \Big\}.$$

Because the cells $\frac{5}{6}(\lambda + V)$ partition $\mathbb{C}$, the sets $\mathscr{C}_0, \dots, \mathscr{C}_{20}$ form a 21-coloring of $\mathbb{C}$.

The congruence condition is equivalent to membership in αL: $p + q\omega \in \alpha L$ if and only if $p - 4q \equiv 0 \pmod{21}$. If $p + q\omega = \alpha(r + s\omega)$, then $p = 4r - s$ and $q = r + 5s$, so $p - 4q = -21s$. Conversely, if $p - 4q = 21k$, then $p + q\omega = \alpha((q + 5k) - k\omega)$.

Lemma 2.1. *The set $\frac{5}{6}(\alpha L + 4H_0)$ is disjoint from the circle $|w| = 2$.*

Proof. The hexagon $\frac{5}{6}(4H_0)$ has circumradius $10/(3\sqrt{3}) < 2$, so the copy centered at 0 lies inside the open disk $|w| < 2$.

Let $0 \neq h = p + q\omega \in L$. Since $|h|^2 = p^2 + pq + q^2$ and $p^2 + pq + q^2 \equiv (p - q)^2 \pmod{3}$, the positive integer $|h|^2$ cannot equal 2. Hence either $|h| = 1$ or $|h| \geq \sqrt{3}$.

If $|h| \geq \sqrt{3}$, then $|\alpha h| \geq \sqrt{63}$, while $4H_0$ has circumradius $4/\sqrt{3}$. Hence $d(\alpha h, 4H_0) \geq \sqrt{63} - 4/\sqrt{3} > 5/2$. If $|h| = 1$, multiplication by h is a rotation preserving L and H_0, so it is enough to consider $h = 1$. We have $\alpha = 9/2 + (\sqrt{3}/2)i$, and the side of $4H_0$ on the line $\operatorname{Re} z = 2$ has endpoints $2 \pm 2i/\sqrt{3}$. The orthogonal projection of α onto this side lies on the segment, so $d(\alpha, 4H_0) = 5/2$.

We have shown that $d(\alpha h, 4H_0) \geq 5/2$ for every nonzero $h \in L$. After scaling by $5/6$, this gives $d(0, \frac{5}{6}(\alpha h + 4H_0)) \geq 25/12 > 2$. Thus every nonzero translate lies outside the disk $|w| \leq 2$, while the copy centered at 0 lies inside it. None of the sets $\frac{5}{6}(\alpha h + 4H_0)$ can therefore meet the circle $|w| = 2$. $\square$

Proof of Theorem 1. Suppose that $P = ABCD$ is monochromatic, with all four vertices in $\mathscr{C}_j$. For each $X \in \{A, B, C, D\}$, write $z_X^2 = \frac{5}{6}(\lambda_X + \varepsilon_X)$, where $\lambda_X = p_X + q_X\omega \in L$ satisfies $p_X - 4q_X \equiv j \pmod{21}$ and $\varepsilon_X \in V \subset H_0$. The alternating sum $\lambda_A - \lambda_B + \lambda_C - \lambda_D$ is congruent to $j - j + j - j = 0$ modulo 21, so it lies in αL. Since H_0 is centrally symmetric and convex, $\varepsilon_A - \varepsilon_B + \varepsilon_C - \varepsilon_D \in 4H_0$. It follows that $\mathscr{I}(P) \in \frac{5}{6}(\alpha L + 4H_0)$. Lemma 2.1 shows that this set is disjoint from the circle $|w| = 2$. Hence $|\mathscr{I}(P)| \neq 2$, and the product of the two adjacent side lengths of P is not 1. $\square$

3. Sharpness of the lattice construction

We next show that no smaller index can work in this construction. Let $M \subset L$ be a finite-index sublattice. Suppose that, after scaling, the copy of $4H_0$ centered at 0 lies inside $|w| < 2$, while every copy centered at a nonzero point of M lies outside the closed disk $|w| \leq 2$. The first condition forces the scaling factor to be less than $\sqrt{3}/2$. The second can then hold only if $d(m, 4H_0) > 4/\sqrt{3}$ for every nonzero $m \in M$.

Proposition 3.1. *If $M \subset L$ has index at most 20, then there exists $0 \neq m \in M$ such that $d(m, 4H_0) \leq 4/\sqrt{3}$.*

Proof. In lattice coordinates, a point $x + y\omega$ lies in $4H_0$ if and only if $|2x + y| \leq 4$, $|x + 2y| \leq 4$, and $|y - x| \leq 4$. Let $Q = 4H_0 - (1 + \omega)/3$. A lattice point $p + q\omega$ lies in Q if and only if $|2p + q + 1| \leq 4$, $|p + 2q + 1| \leq 4$, and $|q - p| \leq 4$. The lattice points in Q are:

q	p
-3	1
-2	$-1, 0, 1, 2$
-1	$-2, -1, 0, 1, 2$
0	$-2, -1, 0, 1$
1	$-3, -2, -1, 0, 1$
2	$-2, -1$

There are 21 lattice points in Q. If $[L : M] \leq 20$, two of them, say x and y, lie in the same coset of M. Their difference $m = x - y$ is a nonzero point of M. Since Q is a translate of $4H_0$, we have $m \in Q - Q = 8H_0$. Hence $m/2 \in 4H_0$ and $|m| \leq 8/\sqrt{3}$. It follows that $d(m, 4H_0) \leq |m - m/2| = |m|/2 \leq 4/\sqrt{3}$. □

Multiplication by $\alpha = 4 + \omega$ scales Euclidean area by $|\alpha|^2 = 21$, so $[L : \alpha L] = 21$. Lemma 2.1 shows that index 21 works, while Proposition 3.1 rules out every index at most 20. Thus 21 is the smallest index for which this sublattice construction can keep the translates away from the circle $|w| = 2$.

Remark 1. Recall that a coloring is Jordan-measurable when the boundary of each color class has Lebesgue measure zero. Kovač's 25-coloring is Jordan-measurable, and so is the coloring constructed here. For our coloring, the boundary of each color class is contained in the inverse image under $z \mapsto z^2$ of the union of the Voronoi-cell boundaries. This union is a countable union of line segments, and the inverse image of each segment is contained in a conic of planar measure zero.

Problem 1. Determine the least integer r for which there exists an r-coloring of $\mathbb{R}^2$ with no monochromatic rectangle of area 1.

Theorem 1 gives $r \leq 21$. Proposition 3.1 shows that the construction of Section 2 cannot produce fewer colors.

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